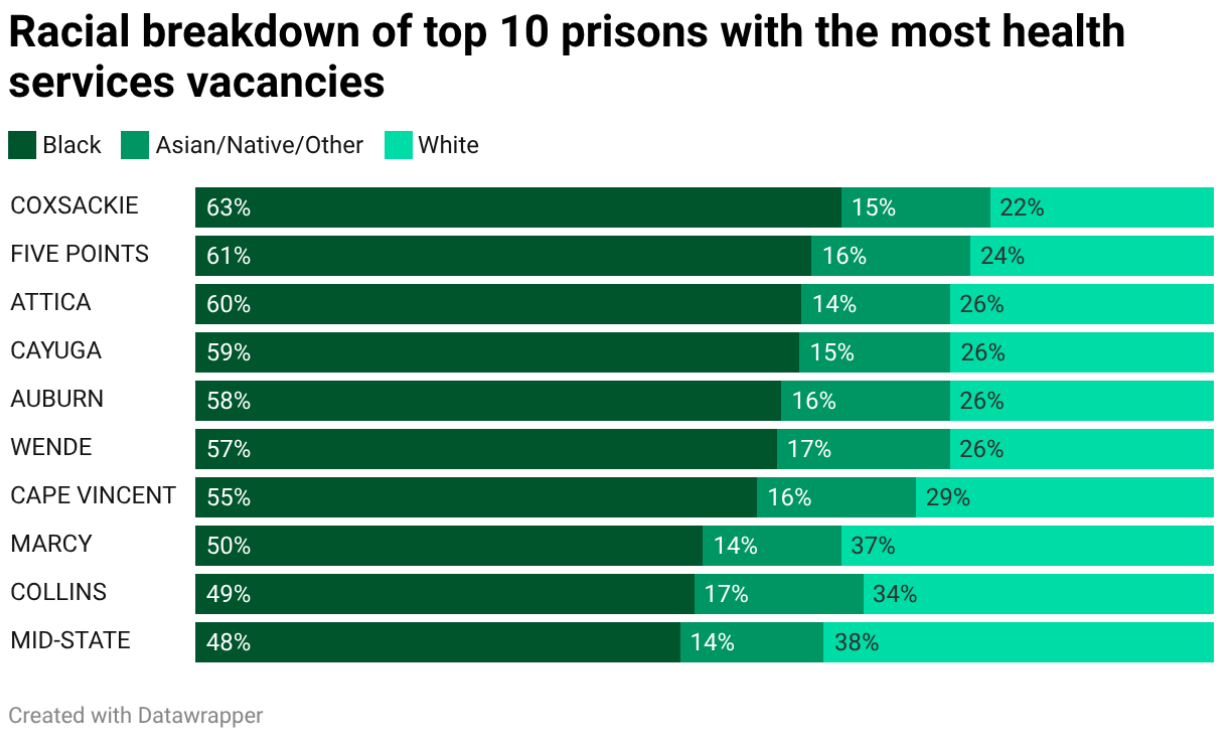
My Methodology



My project involved looking at the correlation between two data sets. The first was a data set showing the filled and budgeted health services positions for New York state prisons and produced by the Correctional Association of New York. This analysis, which I conducted earlier in the year for my capstone, showed that many prisons had a significant amount of vacancies, with the fill rate hovering at around 70% on average. Now, I wanted to see the demographics that were being affected. Using a different data set from CANY, I was able to filter, sort, and group my data in Jupyter Notebook to find a racial breakdown of each facility's population along with its security level (max, min, medium).

I imported my new population data to the Google Sheet containing staffing data. Using the X-lookup function I was able to match facilities with the fill rate, the number of filled positions divided by the number of budgeted positions. That way I had the facility name, the racial breakdown, and the fill rate all on a single sheet. When making my graphs, I had to combine Native and Asian demographics with the categories Other and Unknown because otherwise they would not be visible on the map. I also calculated the percent of each race in a facility, since they can vary greatly in population size. I did something similar using Maximum and minimum security prisons.

My conclusion is that Black individuals were more likely to be affected by the staffing shortages in prisons, even taking into consideration that they make up a bigger portion of the prison

population in general. The Black population in New York maximum security prisons is 6% higher than in medium security prisons. Maximum security prisons are also slightly more likely to be affected by staffing shortages. Across New York's 38 maximum and medium security state prisons, the average Black population is around 57%. Among the top ten, five have black populations above average and the top two— Coxsackie and Five Points—were in the top five prisons with the largest black populations.